III B.Tech II Semester (E.C.E)

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23A04605c	**SATELLITE COMMUNICATIONS**	3	0	0	3

Course Objectives:

1. To learn the principles of orbital mechanics& satellite launch system with performance parameters.
2. To describe the elements of communication satellite design for matching reliability.
3. To know the working concepts of various multiple access techniques and Onboard processing.
4. To analyze the satellite links design with communication links.
5. To evaluate the working of earth station design with satellite broadcasting.

Course Outcomes:

At the end of the course, the students will be able to

1. Learn the principles of orbital mechanics& satellite launch system with performance parameters.
2. Describe the elements of communication satellite design for matching reliability.
3. Gain knowledge on various multiple access techniques and Onboard processing.
4. Analyze the satellite links design with communication links.
5. Evaluate the working of earth station design with satellite broadcasting.

UNIT I

Elements of orbital mechanics. Equations of motion. Tracking and orbit determination. Orbital correction/control. Satellite launch systems. Multistage rocket launchers and their performance

UNIT II

Elements of communication satellite design. Spacecraft subsystems. Reliability considerations. Spacecraft integration.

UNIT III

Multiple access techniques. FDMA, TDMA,CDMA. Random access techniques. Satellite onboard processing.

UNIT IV

Satellite link design: Performance requirements and standards. Design of satellite links – DOMSAT, INSAT, INTELSAT and INMARSAT. Satellite - based personal communication. links.

UNIT V

Earth station design. Configurations. Antenna and tracking systems. Satellite broadcasting.

Textbooks:

D. Roddy, Satellite Communication (4/e), McGraw- Hill, 2009.

T. Pratt & C.W. Bostain, Satellite Communication, Wiley 2000.

References:

B.N. Agrawal, Design of Geosynchrons Spacecraft, Prentice- Hall,1986.